



W1-2-60-1-6

**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY**

University Examinations 2024/2025

SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN COMPUTER TECHNOLOGY

BIT 2111: COMPUTER AIDED DESIGN

DATE: DECEMBER 2024

TIME: 2 HOURS

INSTRUCTIONS: Answer Question **ONE** and any other **TWO** Questions

QUESTION ONE

- (a) Define CAD and explain its role in modern design and manufacturing. **[4 Marks]**
- (b) Discuss the evolution of CAD software from 2D drafting tools to 3D parametric modeling software. How has this evolution changed the way designers and engineers work? **[8 Marks]**
- (c) Evaluate the impact of CAD on industries such as architecture, mechanical engineering, and product design. How has CAD improved efficiency and innovation in these fields? **[8 Marks]**
- (d) You are tasked with designing a mechanical part that requires the use of holes with specific diameters. How would you use CAD software to ensure the holes are uniformly placed on a circular pattern? Detail the steps and features you would use. **[10 Marks]**

QUESTION TWO

- (a) Explain the role of geometric models in CAD and how they help in the design process. **[4 Marks]**
- (b) Discuss the evolution of Computer-Aided Design technology and how it has transformed industries such as architecture and engineering. **[6 Marks]**

- (c) A team is tasked with designing a gear system for a small robot. Using CAD, outline the design process from concept to final assembly. Discuss the importance of each stage, including concept sketching, 3D modeling, assembly design, and prototyping. Explain how collaboration and data management play a role in such a project. **[10 Marks]**

QUESTION THREE

- (a) Briefly explain how the coordinate system is utilized in positioning objects within a CAD drawing. **[4 Marks]**
- (b) What are the different types of dimensions used in CAD? Provide examples of linear, angular, and radial dimensions. **[8 Marks]**
- (c) Discuss the concept of data representation in CAD. How does it impact the accuracy, scalability, and modification of CAD models? **[8 Marks]**

QUESTION FOUR

- (a) Explain the significance of layers and hatch patterns in CAD drawings, and how they improve the clarity and functionality of a drawing. **[4 Marks]**
- (b) Compare and contrast the following types of geometric objects used in CAD: lines, arcs, circles, and polygons. Provide real-world examples of where each might be used in a technical drawing. **[8 Marks]**
- (c) Outline how to create a CAD drawing of a room layout, ensuring you use rectangles, polylines, and hatch patterns for flooring. Add doors and windows as blocks and include proper dimensioning. **[8 Marks]**